Lecture 29.06: SVM Classifier Implementation (`sklearn.svm.SVC`) [Hinglish]









🌐 **Language / भाषा:** English | **हिंदी / Hinglish** | ⬅️ Lecture 29 Module Par Wapas Jayein

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📋 Is Topic Mein Hum Kya Seekhenge?

- `sklearn.svm.SVC` class ka end-to-end pipeline implementation.
- Feature scaling kyu mandatory hai: SVM distance metric par kaam karta hai, isiliye `StandardScaler` zaroori hai.
- Attributes inspection: `clf.support_vectors_` , `clf.n_support_` , `clf.support_` .
- Model evaluation: Confusion matrix, Precision, Recall, F1-score, aur ROC-AUC.

🔍 Har Ek Point Ka Simple & Detailed Explanation

- **Feature Scaling Requirement**: SVM points ke beech Euclidean distance calculate karta hai. Agar ek feature Age (20-60) ho aur doosra Salary (20,000-2,00,000) ho, to Salary dominant ho jayegi aur Age ko neglect kar degi. Hamesha `StandardScaler` lagana zaroori hai!
- **Model Pipeline**: `StandardScaler() -> SVC(kernel='rbf', C=1.0, gamma='scale') -> fit() -> predict()` .
- **Support Vectors Extraction**: Training ke baad `clf.support_vectors_` hume exactly bata deta hai ki konse samples boundary ko define kar rahe hain.
- **Probability Estimates**: Default SVC margin distances dega (`decision_function`). Agar probabilities chahiye hon, to `SVC(probability=True)` set karke `predict_proba()` use kiya jata hai.

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

For an observation $\mathbf{x}$, the continuous signed distance to the margin is computed via the decision function:

$$f(\mathbf{x}) = \sum_{i \in \text{SVs}} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

The predicted discrete class label corresponds to:

$$\hat{y} = \text{sign}(f(\mathbf{x})) = \begin{cases} +1 & \text{if } f(\mathbf{x}) \geq 0 \\ -1 & \text{if } f(\mathbf{x}) < 0 \end{cases}$$

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4. Structured Implementation & Execution

```
import numpy as np
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, roc_auc_score
from sklearn.datasets import load_breast_cancer

def execute_complete_svc_pipeline() -> dict:
    """
    Executes a complete end-to-end Support Vector Classifier workflow on real medi

    Returns:
    --------
    dict
        Evaluation metrics including accuracy, ROC-AUC, and support vector statist
    """
    data = load_breast_cancer()
    X, y = data.data, data.target

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, random_state=42, stratify=y
```

```
    )

    # 1. Feature scaling (Crucial for SVM)
    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    # 2. Fit SVC model
    model = SVC(kernel='rbf', C=1.0, probability=True, random_state=42)
    model.fit(X_train_scaled, y_train)

    # 3. Predict & evaluate
    y_pred = model.predict(X_test_scaled)
    y_prob = model.predict_proba(X_test_scaled)[:, 1]

    train_acc = model.score(X_train_scaled, y_train)
    test_acc = model.score(X_test_scaled, y_test)
    auc_score = roc_auc_score(y_test, y_prob)

    return {
        "dataset_name": "Breast Cancer Diagnostic",
        "train_accuracy": float(train_acc),
        "test_accuracy": float(test_acc),
        "roc_auc_score": float(auc_score),
        "total_support_vectors": int(np.sum(model.n_support_)),
        "class_support_breakdown": model.n_support_.tolist()
    }

# Driver test
svc_results = execute_complete_svc_pipeline()
print("SVC Complete Pipeline Results:")
for k, v in svc_results.items():
    print(f"  {k}: {v}")
```

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5. Mukhya Batein (Key Takeaways) & Summary

- Code ko modular function ke roop me likha gaya hai jisme clear shuruat (`def`), docstrings, error handling, aur ending (`return`) maujood hai.
- Video lectures me ma'am dwara sikhaye gaye exact datasets (Diabetes, concentric circles, blobs) aur syntax parameters ko exact follow kiya gaya hai.